NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Assignment:-02

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Roll No: …………………… SP21379

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1. Rule-Based Expert System

Implementation (in Python):

python

Copy code

```
def diagnose_car_issue():
    print("Car Engine Diagnostic Expert System")
    engine_start = input("Does the engine start? (Yes/No): ").strip().lower()
    fuel_tank_full = input("Is the fuel tank full? (Yes/No): ").strip().lower()
    spark_at_plugs = input("Is there a spark at the spark plugs? (Yes/No): ").strip().lower()

    if engine_start == "no":
        if fuel_tank_full == "no":
            print("Diagnosis: Out of fuel.")
        elif spark_at_plugs == "no":
            print("Diagnosis: No spark.")
        else:
            print("Diagnosis: Faulty starter motor.")
    else:
        print("No issues detected with the engine.")

# Run the system
diagnose_car_issue()
```

Rules and Decision-Making Process:

- Rule 1: If engine_start = No AND fuel_tank_full = No, then diagnosis = "Out of fuel".
- Rule 2: If engine_start = No AND spark_at_plugs = No, then diagnosis = "No spark".
- Rule 3: If engine_start = No AND fuel_tank_full = Yes AND spark_at_plugs = Yes, then diagnosis = "Faulty starter motor".
- Rule 4: If engine_start = Yes, then diagnosis = "No issues detected".

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2. Forward vs. Backward Chaining

Comparison:

- **Forward Chaining:**
  Starts with known facts and applies rules to infer new facts until a conclusion is reached.
  Example:
  - Known Symptoms: Fever, cough, fatigue.
  - Rules:
    - Rule 1: If fever AND cough, infer "Flu".
    - Rule 2: If fever AND cough AND fatigue, infer "COVID-19".
  - Process: Starting from symptoms, rules are applied sequentially to deduce the condition.
- **Backward Chaining:**
  Starts with a hypothesis (goal) and works backward to confirm it by checking if known facts support it.
  Example:
  - Hypothesis: "Is it COVID-19?"
  - Rules:
    - Rule 1: If fever AND cough AND fatigue, confirm "COVID-19".
    - Process: Check symptoms to confirm or deny the hypothesis.

Efficiency:

- **Forward Chaining:** More efficient when all possible diagnoses are needed because it explores all rule consequences.
- **Backward Chaining:** More efficient when confirming a specific diagnosis since it focuses only on rules relevant to the hypothesis.

Recommendation: Use backward chaining for targeted diagnosis (e.g., checking for COVID-19) and forward chaining for exploring multiple possible conditions.

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3. Expert System Architecture for Diagnosing Network Issues

Components:

1. **Knowledge Base:**
   - Contains rules and facts about network issues.
   - Example Rules:
     - Rule 1: If no internet AND Wi-Fi disconnected, suggest "Reconnect Wi-Fi".
     - Rule 2: If slow speeds AND high bandwidth usage, suggest "Limit bandwidth use".

2. **Inference Engine:**
   - Evaluates user inputs against the knowledge base to deduce a solution.
   - Uses chaining (forward/backward) to draw conclusions.
3. **User Interface:**
   - Allows users to input symptoms and receive suggestions.

How the Inference Engine Works:

- **Step 1:** Gather facts (user inputs, e.g., "no internet").
- **Step 2:** Match inputs against rules in the knowledge base.
- **Step 3:** Infer a solution (e.g., "Reconnect Wi-Fi").
- **Step 4:** Provide step-by-step troubleshooting instructions.

Knowledge Base Structure:

- Facts: Predefined states like "no internet", "slow speeds".
- Rules: Logical if-then constructs to deduce problems and solutions.

Example Workflow:

1. User Input: "My internet is not working."
2. System Checks:
   - Is the router powered on? (Yes/No)
   - Is the device connected to Wi-Fi? (Yes/No)
   - Is there high latency? (Yes/No)
3. Diagnosis: Based on responses, the system suggests steps to fix the issue.